

AES REGISTER SET FOR AXI SLAVE INTERFACE

AES REGISTER SET

Register Width: 32 bits

The AES IP consists of two blocks:

1. AES Normal Cipher
2. AES Inverse Cipher

Each 128-bit AES key, input and output is divided into four 32-bit registers.

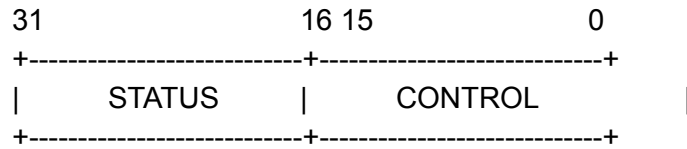
1. AES NORMAL CIPHER REGISTER SET

Base Address: AES_CIPHER_BASE

Register Name	Offset	Width	Description
CIPHER_CSR	0x00	32-bit	Control/Status Register
CIPHER_KEY0	0x04	32-bit	Key [31:0]
CIPHER_KEY1	0x08	32-bit	Key [63:32]
CIPHER_KEY2	0x0C	32-bit	Key [95:64]
CIPHER_KEY3	0x10	32-bit	Key [127:96]
CIPHER_TIN0	0x14	32-bit	Text Input [31:0]
CIPHER_TIN1	0x18	32-bit	Text Input [63:32]
CIPHER_TIN2	0x1C	32-bit	Text Input [95:64]
CIPHER_TIN3	0x20	32-bit	Text Input [127:96]
CIPHER_TOUT0	0x24	32-bit	Text Output [31:0]
CIPHER_TOUT1	0x28	32-bit	Text Output [63:32]
CIPHER_TOUT2	0x2C	32-bit	Text Output [95:64]
CIPHER_TOUT3	0x30	32-bit	Text Output [127:96]

CIPHER_CSR - CONTROL / STATUS REGISTER

Register Width: 32 bits



CONTROL [15:0]:

Bit 0 : LD
Load/Start AES encryption operation.

Bits 1-15 : Reserved

STATUS [31:16]:

Bit 16 : Reserved
Bit 17 : Reserved
Bit 18 : Reserved
Bit 19 : Reserved
Bit 20 : Reserved
Bit 21 : Reserved
Bit 22 : Reserved
Bit 23 : Reserved
Bit 24 : Reserved
Bit 25 : Reserved
Bit 26 : Reserved
Bit 27 : Reserved
Bit 28 : Reserved
Bit 29 : Reserved
Bit 30 : Reserved
Bit 31 : DONE
Indicates completion of AES encryption operation.

NORMAL CIPHER_CSR SUMMARY

CSR[31] = DONE
 CSR[30:16] = Reserved
 CSR[15:1] = Reserved
 CSR[0] = LD

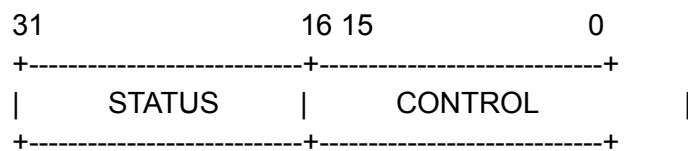
2. AES INVERSE CIPHER REGISTER SET

Base Address: AES_INV_CIPHER_BASE

Register Name	Offset	Width	Description
INV_CIPHER_CSR	0x00	32-bit	Control/Status Register
INV_KEY0	0x04	32-bit	Key [31:0]
INV_KEY1	0x08	32-bit	Key [63:32]
INV_KEY2	0x0C	32-bit	Key [95:64]
INV_KEY3	0x10	32-bit	Key [127:96]
INV_TIN0	0x14	32-bit	Cipher Input [31:0]
INV_TIN1	0x18	32-bit	Cipher Input [63:32]
INV_TIN2	0x1C	32-bit	Cipher Input [95:64]
INV_TIN3	0x20	32-bit	Cipher Input [127:96]
INV_TOUT0	0x24	32-bit	Plaintext Output [31:0]
INV_TOUT1	0x28	32-bit	Plaintext Output [63:32]
INV_TOUT2	0x2C	32-bit	Plaintext Output [95:64]
INV_TOUT3	0x30	32-bit	Plaintext Output [127:96]

INV_CIPHER_CSR - CONTROL / STATUS REGISTER

Register Width: 32 bits



CONTROL [15:0]:

Bit 0 : LD

Load/Start AES inverse cipher operation.

Bit 1 : KLD

Load key into AES inverse cipher.

Bits 2-15 : Reserved

STATUS [31:16]:

Bit 16 : Reserved

Bit 17 : Reserved

Bit 18 : Reserved

Bit 19 : Reserved

Bit 20 : Reserved

Bit 21 : Reserved

Bit 22 : Reserved

Bit 23 : Reserved

Bit 24 : Reserved

Bit 25 : Reserved

Bit 26 : Reserved

Bit 27 : Reserved

Bit 28 : Reserved

Bit 29 : Reserved

Bit 30 : KDONE

Indicates completion of key loading.

Bit 31 : DONE

Indicates completion of AES inverse cipher operation.

INVERSE CIPHER CSR SUMMARY

CSR[31] = DONE

CSR[30] = KDONE

CSR[29:16] = Reserved

CSR[15:2] = Reserved

CSR[1] = KLD

CSR[0] = LD

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3. 128-BIT DATA MAPPING

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All 128-bit AES values are divided into four 32-bit registers.

AES KEY:

KEY[127:0] = {KEY3, KEY2, KEY1, KEY0}

TEXT INPUT:

TEXT_IN[127:0] = {TIN3, TIN2, TIN1, TIN0}

TEXT OUTPUT:

TEXT_OUT[127:0] = {TOUT3, TOUT2, TOUT1, TOUT0}

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4. REGISTER ACCESS SUMMARY

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NORMAL CIPHER:

CIPHER_CSR	0x00	32-bit
CIPHER_KEY0	0x04	32-bit
CIPHER_KEY1	0x08	32-bit
CIPHER_KEY2	0x0C	32-bit
CIPHER_KEY3	0x10	32-bit
CIPHER_TIN0	0x14	32-bit
CIPHER_TIN1	0x18	32-bit
CIPHER_TIN2	0x1C	32-bit
CIPHER_TIN3	0x20	32-bit
CIPHER_TOUT0	0x24	32-bit
CIPHER_TOUT1	0x28	32-bit
CIPHER_TOUT2	0x2C	32-bit

CIPHER_TOUT3 0x30 32-bit

INVERSE CIPHER:

INV_CIPHER_CSR	0x00	32-bit
INV_KEY0	0x04	32-bit
INV_KEY1	0x08	32-bit
INV_KEY2	0x0C	32-bit
INV_KEY3	0x10	32-bit
INV_TIN0	0x14	32-bit
INV_TIN1	0x18	32-bit
INV_TIN2	0x1C	32-bit
INV_TIN3	0x20	32-bit
INV_TOUT0	0x24	32-bit
INV_TOUT1	0x28	32-bit
INV_TOUT2	0x2C	32-bit
INV_TOUT3	0x30	32-bit

5. REGISTER COUNT AND ADDRESS SPACE

Normal Cipher:

13 registers
Register offsets: 0x00 to 0x30
Register width: 32 bits
Total register space: 0x34 bytes

Inverse Cipher:

13 registers
Register offsets: 0x00 to 0x30
Register width: 32 bits
Total register space: 0x34 bytes

6. CSR BIT SUMMARY

NORMAL CIPHER CSR:

CONTROL [15:0]

Bit 0 = LD

Bits 15:1 = Reserved

STATUS [31:16]

Bits 31:16 = Status field

Bit 31 = DONE

Bits 30:16 = Reserved

INVERSE CIPHER CSR:

CONTROL [15:0]

Bit 0 = LD

Bit 1 = KLD

Bits 15:2 = Reserved

STATUS [31:16]

Bits 31:16 = Status field

Bit 31 = DONE

Bit 30 = KDONE

Bits 29:16 = Reserved

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END OF REGISTER SET

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